Post-Quantum Cryptography

Lattices II

Learning With Errors, ML-KEM and ML-DSA

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Outline

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2 Encryption from LWE

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Where we are

Previous lecture gave us SIS: find a *short* solution to $\mathbf{A}\mathbf{z} \equiv \mathbf{0}$. SIS yields collision-resistant hashing and signatures, but not encryption — there is no trapdoor for decryption.

This lecture

Learning With Errors (LWE) is, informally, the "dual" of SIS. It supports encryption directly, has its own worst-case hardness reduction, and is the basis of both deployed lattice standards.

We proceed in four steps:

1. LWE: definition, intuition, and why the noise is essential.
2. Regev's encryption scheme and its security proof.
3. Ring- and Module-LWE: the algebraic structure that makes it practical.
4. ML-KEM and ML-DSA as actually specified.

Solving linear equations — with and without noise

Suppose $\mathbf{s} \in \mathbb{Z}_q^n$ is secret and you are given samples $(\mathbf{a}_i,\ b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle)$ with $\mathbf{a}_i$ uniform.

Without noise: collect n samples, form a matrix, run Gaussian elimination. Recovered in $\mathcal{O}(n^3)$. Completely insecure.

Now perturb each equation by a small error e_i:

$$b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q.$$

Everything changes

Gaussian elimination amplifies the errors catastrophically: taking linear combinations of $\mathcal{O}(n)$ equations multiplies the noise by large coefficients, and modulo q the result is indistinguishable from uniform. No efficient algorithm is known for recovering $\mathbf{s}$.

This is the entire idea. The problem is elementary to state, requires no algebraic number theory, and can be explained to a second-year student in five minutes — which is one reason lattice cryptography became dominant.

The LWE problem

Definition 1 (LWE distribution)

For $\mathbf{s} \in \mathbb{Z}_q^n$ and an error distribution χ over $\mathbb{Z}$, the distribution $A_{\mathbf{s},\chi}$ outputs

$$(\mathbf{a},\ b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q), \qquad \mathbf{a} \leftarrow \mathbb{Z}_q^n \text{ uniform},\ e \leftarrow \chi.$$

Definition 2 (Search-LWE and Decision-LWE)

Given m samples from $A_{\mathbf{s},\chi}$ with $\mathbf{s}$ uniform, *search*-LWE asks you to recover $\mathbf{s}$; *decision*-LWE asks you to distinguish those samples from m uniform samples of $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

In matrix form: given $(\mathbf{A},\ \mathbf{b} = \mathbf{A}^T\mathbf{s} + \mathbf{e})$ with $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, recover $\mathbf{s}$, or distinguish $\mathbf{b}$ from uniform.

Theorem 1 (Search–decision equivalence [15])

For prime $q = \mathrm{poly}(n)$ the two versions are equivalent up to polynomial factors.

LWE, geometrically

Consider the lattice

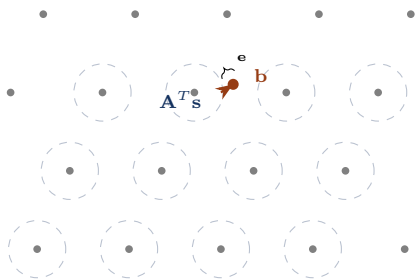
$$\mathcal{L}(\mathbf{A}) = \{\mathbf{A}^T\mathbf{s} \bmod q : \mathbf{s} \in \mathbb{Z}_q^n\} \subseteq \mathbb{Z}_q^m.$$

An LWE sample vector $\mathbf{b} = \mathbf{A}^T\mathbf{s} + \mathbf{e}$ is a lattice point plus a *small* perturbation.

So search-LWE is exactly Bounded Distance Decoding on $\mathcal{L}(\mathbf{A})$: find the lattice point near a given target, knowing the target is close.

Decision-LWE asks whether the target is close to the lattice at all, or just a uniformly random point of $\mathbb{Z}_q^m$.

The error magnitude relative to λ_1 controls hardness: too much noise and decryption fails, too little and the problem becomes easy.



Lattice points with their decoding radii. A sample lands inside one ball; the secret is the centre. Random points miss the balls entirely.

Choosing the error distribution

The noise must be small enough to decode, large enough to hide the secret.

Discrete Gaussian $D_{\mathbb{Z},\sigma}$

$\Pr[X = x] \propto e^{-\pi x^2/\sigma^2}$.

Used in the theory: the worst-case reductions are proved for Gaussian noise, and Gaussians behave well under the linear operations the proofs need.

Hard to sample in constant time. Naive samplers branch on secret data and leak through timing — a real source of implementation vulnerabilities.

Centred binomial B_η

Sample $a_1, \ldots, a_\eta, b_1, \ldots, b_\eta \leftarrow \{0,1\}$ and output $\sum_i (a_i - b_i)$.

Range $[-\eta, \eta]$, variance $\eta/2$, and *trivially constant-time*: just count bits.

Used by ML-KEM ($\eta \in \{2,3\}$) and, in a uniform variant, by ML-DSA. The distribution differs from a Gaussian, but the security loss is small and well quantified [2].

Key idea

A recurring theme in post-quantum engineering: the distribution that is easiest to *prove* things about is not the one that is safest to *implement*. Standards chose implementability.

Regev's worst-case hardness theorem

Theorem 2 ([15])

Let $q = \mathrm{poly}(n)$ and let χ be a discrete Gaussian with $\sigma \geq 2\sqrt{n}$. If there is an efficient algorithm solving decision-LWE$_{n,q,\chi}$, then there is an efficient quantum algorithm solving SIVP_γ and GapSVP_γ on every n-dimensional lattice, for $\gamma = \tilde{\mathcal{O}}(n/\alpha)$.



Read the statement carefully

The reduction itself is *quantum*. It says: if LWE is easy, then lattice problems are easy *for a quantum computer*. Since we assume quantum computers exist in our threat model, this is exactly the guarantee we want — but it is a subtler statement than the classical SIS reduction, and students often misstate it. Later work gave classical reductions for some parameter ranges.

Regev's encryption scheme

Parameters

Dimension n, modulus q, samples m, error distribution χ. Message space $\{0,1\}$.

KeyGen. Sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e} \leftarrow \chi^m$. Set

$$\mathbf{b}^T = \mathbf{s}^T\mathbf{A} + \mathbf{e}^T, \qquad pk = (\mathbf{A}, \mathbf{b}), \qquad sk = \mathbf{s}.$$

Encrypt $\mu \in \{0,1\}$. Sample $\mathbf{r} \leftarrow \{0,1\}^m$ and output

$$\mathbf{c}_1 = \mathbf{A}\mathbf{r}, \qquad c_2 = \langle \mathbf{b}, \mathbf{r} \rangle + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor.$$

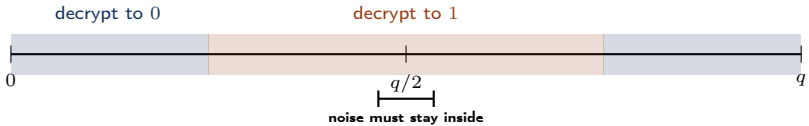
Decrypt. Compute $c_2 - \langle \mathbf{s}, \mathbf{c}_1 \rangle$ and output 0 if the result is closer to 0 than to $\lfloor q/2 \rfloor$, else 1.

Why decryption works

Expand:

$$c_2 - \langle \mathbf{s}, \mathbf{c}_1 \rangle = \underbrace{\langle \mathbf{b}, \mathbf{r} \rangle - \langle \mathbf{s}, \mathbf{A}\mathbf{r} \rangle}_{= \langle \mathbf{e}, \mathbf{r} \rangle} + \mu \left\lfloor \frac{q}{2} \right\rfloor = \mu \left\lfloor \frac{q}{2} \right\rfloor + \langle \mathbf{e}, \mathbf{r} \rangle.$$

Since $\mathbf{r}$ is binary and $\mathbf{e}$ is small, the term $\langle \mathbf{e}, \mathbf{r} \rangle$ is a sum of at most m small errors — much smaller than $q/4$ with high probability. Rounding recovers μ.



Decryption failures

The noise bound holds only *with high probability*. A failure probability that is negligible but nonzero is unavoidable, and it is a genuine attack surface: an adversary who can trigger and observe failures learns about the secret [3]. Standards target failure probabilities below 2^{-138}.

Security of Regev encryption

Theorem 3

If decision-LWE$_{n,q,\chi}$ is hard, Regev's scheme is IND-CPA secure.

Proof idea — a hybrid argument.

Game 0 The real scheme.

Game 1 Replace $\mathbf{b} = \mathbf{s}^T\mathbf{A} + \mathbf{e}^T$ by a *uniform* $\mathbf{b}$. Distinguishing Game 0 from Game 1 is exactly solving decision-LWE, so the gap is negligible.

Game 2 Now $(\mathbf{A}, \mathbf{b})$ is uniform. By the leftover hash lemma, $(\mathbf{Ar}, \langle \mathbf{b}, \mathbf{r} \rangle)$ for random binary $\mathbf{r}$ is statistically close to uniform, provided m is large enough. Hence the ciphertext carries no information about μ.

The adversary's advantage in Game 2 is 0; the total gap is negligible. □ □

Note the shape: this is the reduction template from Lecture 1, instantiated. Each game changes one thing, and each change is justified either by a hardness assumption or by a statistical argument.

The efficiency problem

Regev's scheme is secure and simple. It is also unusable at scale.

Sizes

The public key contains $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ with $m \approx n \log q$: that is $\mathcal{O}(n^2 \log q)$ bits.

For $n = 256$ and $q \approx 2^{12}$ this is already megabytes, and the scheme encrypts one bit per ciphertext.

The fix

Impose algebraic structure so that $\mathbf{A}$ is determined by $\mathcal{O}(n)$ elements rather than $\mathcal{O}(n^2)$, and so that multiplication is fast.

Replace vectors over $\mathbb{Z}_q$ by elements of a polynomial ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$.

The trade-off, stated honestly

Structure buys efficiency and costs generality. The worst-case hardness now refers to *ideal* or *module* lattices, a restricted class. No attack exploiting this structure is known for the parameters in use — but the security assumption is strictly stronger than plain LWE.

The ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$

Take n a power of two and work with polynomials of degree $< n$, reduced modulo $x^n + 1$ and coefficient-wise modulo q.

The reduction rule is simply $x^n \equiv -1$: a *negacyclic* convolution.

Example 1

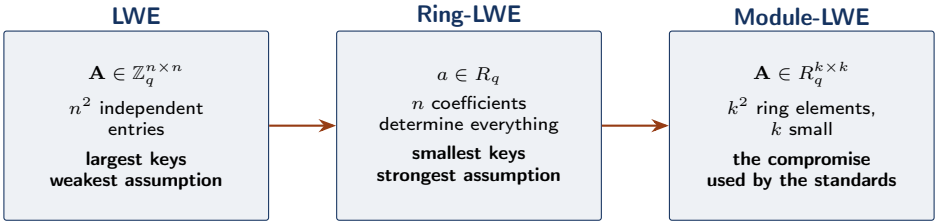
With $n = 4$, $x^3 \cdot x^2 = x^5 = x \cdot x^4 \equiv -x$.

Why this ring

- $x^n + 1$ is the $2n$-th cyclotomic polynomial for n a power of two — irreducible over $\mathbb{Q}$, giving a well-behaved ring.
- If $q \equiv 1 \bmod 2n$, then $x^n + 1$ splits completely modulo q, so by the Chinese remainder theorem $R_q \cong \mathbb{Z}_q^n$ as rings. Multiplication becomes coordinate-wise.
- That isomorphism is computed by the *number-theoretic transform*, an FFT over $\mathbb{Z}_q$, in $\mathcal{O}(n \log n)$ operations.

ML-KEM uses $n = 256$, $q = 3329 = 13 \cdot 256 + 1$, chosen precisely so the NTT applies.

Plain, Ring, and Module LWE



Module-LWE [8] interpolates: take $k \times k$ matrices over R_q. Setting $k = 1$ gives Ring-LWE; letting the ring be trivial gives plain LWE.

Key idea

Module-LWE lets one change the security level by changing k alone, keeping the ring — and therefore the entire NTT implementation — fixed. That is why ML-KEM-512, -768 and -1024 share one codebase and differ only in $k = 2, 3, 4$. This is an engineering argument, and it decided the standard.

From encryption to key encapsulation

In practice we do not encrypt messages with lattice schemes. We establish a symmetric key, then use AES. The right abstraction is a KEM.

Definition 3 (Key Encapsulation Mechanism)

A triple of algorithms:

- $\mathrm{KeyGen}() \to (pk, sk)$;
- $\mathrm{Encaps}(pk) \to (c, K)$ — outputs a ciphertext and a shared key;
- $\mathrm{Decaps}(sk, c) \to K$.

Why a KEM rather than encryption

The sender does not choose the key — it is derived from randomness inside Encaps. This removes a class of misuse, simplifies the security definition, and matches how TLS actually establishes traffic keys.

We need IND-CCA security: the adversary may query a decapsulation oracle. Regev's scheme is only IND-CPA, so a transformation is required.

The Fujisaki–Okamoto transform

The idea

Make encryption *deterministic* given the message, so the decryptor can re-encrypt and check.

Encaps(pk):

1. Sample random m; compute $(\bar{K}, r) = G(m, H(pk))$.
2. $c = \mathrm{Enc}(pk, m; r)$ — encrypt *using r as the randomness*.
3. Output $(c,\ K = \mathrm{KDF}(\bar{K}, c))$.

Decaps(sk, c):

1. $m' = \mathrm{Dec}(sk, c)$; recompute $(\bar{K}', r') = G(m', H(pk))$ and re-encrypt $c' = \mathrm{Enc}(pk, m'; r')$.
2. If $c' = c$ output $\mathrm{KDF}(\bar{K}', c)$; otherwise output a pseudorandom value derived from a secret seed and c.

Implicit rejection

Step 3 must not reveal that decapsulation failed. Returning an error, or returning early, hands the adversary an oracle — the most common implementation error in lattice KEMs, and a timing side channel by default.

Security is proved in the (quantum) random oracle model [1, 6, 7].

Why the Fujisaki–Okamoto transform

The idea

to upgrade a passively secure public-key encryption (PKE) or key encapsulation mechanism (KEM) — one secure only against chosen-plaintext attacks (CPA)—into a robust scheme secure against adaptive chosen-ciphertext attacks (CCA)

1. Eliminating Malleability: Basic encryption schemes are often malleable, meaning an attacker can modify a ciphertext and produce a predictably altered plaintext or manipulate the system. CCA security ensures that any tampering with a ciphertext invalidates it completely.
2. Adding Integrity via Re-Encryption: The transform forces the decryption/decapsulation process to perform a re-encryption check. It decrypts the incoming ciphertext, re-encrypts the resulting message, and compares the newly generated ciphertext with the original one. If they do not match, the system rejects the input, preventing attackers from probing the secret key
3. Post-Quantum Standard: It serves as the primary "recipe" used to secure post-quantum KEM candidates (like FrodoKEM) standardized by NIST. Many lattice-based and code-based cryptographic primitives suffer from small decryption failure probabilities; modern revisions of the FO transform specifically account for these failures in both the random oracle model (ROM) and quantum random oracle model (QROM).

ML-KEM parameters

	k	public key	ciphertext	secret key	NIST level
ML-KEM-512	2	800 B	768 B	1632 B	1 (AES-128)
ML-KEM-768	3	1184 B	1088 B	2400 B	3 (AES-192)
ML-KEM-1024	4	1568 B	1568 B	3168 B	5 (AES-256)

Common to all three: $n = 256$, $q = 3329$, ring $R_q = \mathbb{Z}_{3329}[x]/(x^{256} + 1)$. Only k and the noise parameters change.

Compression

Ciphertexts are compressed by discarding low-order bits of each coefficient. This adds a small, deliberate rounding error — acceptable because the scheme already tolerates noise, and it reduces ciphertext size by roughly a third.

Comparison

X25519 key exchange: 32-byte public key, 32-byte ciphertext.

ML-KEM-768 is therefore about $35\times$ larger, but still small enough for a TLS handshake. Speed is comparable or better.

ML-KEM-768 is the common deployment choice, and is what hybrid TLS key exchange currently uses in practice [11].

Signatures from lattices: the identification route

Schnorr-style identification, adapted

1. **Commit.** Sample a masking vector $\mathbf{y}$ with small coefficients; compute $\mathbf{w} = \mathbf{A}\mathbf{y}$.
2. **Challenge.** $c = H(\mathbf{w},\ \text{message})$ — a sparse polynomial with ± 1 coefficients (Fiat–Shamir makes this non-interactive).
3. **Respond.** $\mathbf{z} = \mathbf{y} + c\mathbf{s}$, where $\mathbf{s}$ is the secret.
4. **Verify.** Check $\mathbf{A}\mathbf{z} - c\mathbf{t} = \mathbf{w}$ and that $\mathbf{z}$ is short, where $\mathbf{t} = \mathbf{A}\mathbf{s}$ is the public key.

The leak

$\mathbf{z} = \mathbf{y} + c\mathbf{s}$ depends on the secret $\mathbf{s}$. Its distribution is a shifted copy of $\mathbf{y}$'s distribution, and the shift is $c\mathbf{s}$. Many signatures reveal the shift — this is exactly the GGH/NTRU failure from Lecture 3 [13].

Fiat–Shamir with aborts

Lyubashevsky [9] solved this with rejection sampling.

The rule

After computing $\mathbf{z} = \mathbf{y} + c\mathbf{s}$, check whether $\|\mathbf{z}\|_\infty$ exceeds a bound $\gamma_1 - \beta$. If it does, discard everything and restart with fresh $\mathbf{y}$.



Only signatures whose $\mathbf{z}$ falls in the region common to *every* possible shift are output. On that region the output distribution is identical regardless of $\mathbf{s}$, so the signature is provably zero-knowledge: it leaks nothing about the secret.

The cost is repetition — ML-DSA restarts a handful of times on average, which makes signing time variable but the *distribution* of the output key-independent.

ML-DSA parameters

	(k, ℓ)	public key	signature	NIST level
ML-DSA-44	$(4, 4)$	1312 B	2420 B	2
ML-DSA-65	$(6, 5)$	1952 B	3309 B	3
ML-DSA-87	$(8, 7)$	2592 B	4627 B	5

Ring: $R_q = \mathbb{Z}_q[x]/(x^{256} + 1)$ with $q = 8380417 = 2^{23} - 2^{13} + 1$, again chosen for the NTT.

Compared with what it replaces

Ed25519: 32-byte key, 64-byte signature.

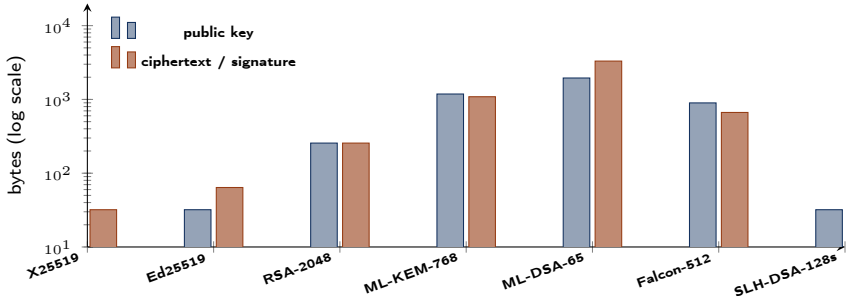
ML-DSA-65: $61\times$ the key, $52\times$ the signature. For certificate chains carrying several signatures, this is the dominant migration cost.

Falcon (FN-DSA), for contrast

Signatures around 666 bytes at level 1 — far smaller. But it requires floating-point Gaussian sampling, which is difficult to implement in constant time and hard to make deterministic across platforms [5].

ML-DSA was standardised first precisely because it is easier to implement correctly.

Putting the sizes in perspective



Three observations for the exercises. SLH-DSA has a tiny key and an enormous signature. Falcon is the most compact lattice signature but the hardest to implement. ML-KEM sits close enough to the classical schemes that TLS absorbs it without redesign — which is why key exchange migrated first.

Summary

- LWE is solving noisy linear equations modulo q. Remove the noise and it is Gaussian elimination; add it and no efficient algorithm is known.
- Search-LWE is Bounded Distance Decoding on a q-ary lattice; Regev gave a quantum worst-case reduction from SIVP.
- Regev's encryption is simple and provably IND-CPA, but has $\mathcal{O}(n^2)$ keys and encrypts one bit.
- Ring- and Module-LWE add algebraic structure, reducing keys to $\mathcal{O}(n)$ and enabling $\mathcal{O}(n \log n)$ multiplication via the NTT, at the cost of a stronger assumption.
- ML-KEM = Module-LWE encryption + Fujisaki–Okamoto, giving IND-CCA.
- ML-DSA = Fiat–Shamir with aborts, where rejection sampling makes the signature distribution independent of the secret key.

Exercises

Exercise 1

Let $q = 17$, $n = 3$, $\mathbf{s} = (2, 5, 11)$. Construct three noiseless samples $(\mathbf{a}_i, \langle \mathbf{a}_i, \mathbf{s} \rangle)$ and recover $\mathbf{s}$ by Gaussian elimination. Then add errors $e_i \in \{-1, 0, 1\}$ and show that the same elimination now yields a wrong answer.

Exercise 2

Verify the decryption identity $c_2 - \langle \mathbf{s}, \mathbf{c}_1 \rangle = \mu \lfloor q/2 \rfloor + \langle \mathbf{e}, \mathbf{r} \rangle$ from the definitions, and derive an explicit condition on $\|\mathbf{e}\|$ and m under which decryption is always correct.

Exercise 3

In $R_q = \mathbb{Z}_{17}[x]/(x^4 + 1)$, compute $(1 + 2x + 3x^3) \cdot (2 + x^2)$. Show each reduction step using $x^4 \equiv -1$.

Exercise 4

Explain why $q = 3329$ and $n = 256$ satisfy $q \equiv 1 \bmod 2n$, and state what would go wrong with the ML-KEM implementation if this condition failed.

Exercises (continued)

Exercise 5

Give a concrete attack against a KEM whose Decaps returns an explicit error on re-encryption failure, and explain how implicit rejection prevents it.

Exercise 6

In Fiat–Shamir with aborts, suppose the rejection bound were removed entirely. Describe how an adversary collecting many signatures could recover $\mathbf{s}$, and relate this to the attack of Nguyen and Regev on GGH signatures.

Exercise 7 (Discussion)

Module-LWE assumes more structure than plain LWE and is therefore a strictly stronger assumption. Argue both sides of the question of whether standardising Module-LWE was the right decision, referring to key sizes, implementation maturity, and the risk of a structure-exploiting attack.

References I

[1] Dan Boneh, Özgür Dagdelen, Marc Fischlin, Anja Lehmann, Christian Schaffner, and Mark Zhandry. Random oracles in a quantum world. In *Advances in Cryptology – ASIACRYPT 2011*, pages 41–69, 2011.

[2] Joppe Bos, Léo Ducas, Eike Kiltz, Tancrède Lepoint, Vadim Lyubashevsky, John M. Schanck, Peter Schwabe, Gregor Seiler, and Damien Stehlé. CRYSTALS-Kyber: A CCA-secure module-lattice-based KEM. In *IEEE European Symposium on Security and Privacy (EuroS&P)*, pages 353–367, 2018.

[3] Jan-Pieter D'Anvers, Qian Guo, Thomas Johansson, Alexander Nilsson, Frederik Vercauteren, and Ingrid Verbauwhede. Decryption failure attacks on IND-CCA secure lattice-based schemes. In *Public-Key Cryptography – PKC 2019*, pages 565–598, 2019.

[4] Léo Ducas, Eike Kiltz, Tancrède Lepoint, Vadim Lyubashevsky, Peter Schwabe, Gregor Seiler, and Damien Stehlé. CRYSTALS-Dilithium: A lattice-based digital signature scheme. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2018(1):238–268, 2018.

[5] Pierre-Alain Fouque, Jeffrey Hoffstein, Paul Kirchner, Vadim Lyubashevsky, Thomas Pornin, Thomas Prest, Thomas Ricosset, Gregor Seiler, William Whyte, and Zhenfei Zhang. Falcon: Fast-fourier lattice-based compact signatures over NTRU. In *NIST Post-Quantum Cryptography Standardization Round 1 Submission*, 2018.

[6] Eiichiro Fujisaki and Tatsuaki Okamoto. Secure integration of asymmetric and symmetric encryption schemes. In *Advances in Cryptology – CRYPTO '99*, pages 537–554, 1999.

[7] Dennis Hofheinz, Kathrin Hövelmanns, and Eike Kiltz. A modular analysis of the Fujisaki-Okamoto transformation. In *Theory of Cryptography Conference (TCC)*, pages 341–371, 2017.

References II

[8] Adeline Langlois and Damien Stehlé. Worst-case to average-case reductions for module lattices. *Designs, Codes and Cryptography*, 75(3):565–599, 2015.

[9] Vadim Lyubashevsky. Fiat-shamir with aborts: Applications to lattice and factoring-based signatures. In *Advances in Cryptology – ASIACRYPT 2009*, pages 598–616, 2009.

[10] Vadim Lyubashevsky, Chris Peikert, and Oded Regev. On ideal lattices and learning with errors over rings. *Journal of the ACM*, 60(6):1–35, 2013. Preliminary version in EUROCRYPT 2010.

[11] National Institute of Standards and Technology. Module-lattice-based key-encapsulation mechanism standard. Technical Report FIPS 203, U.S. Department of Commerce, 2024.

[12] National Institute of Standards and Technology. Module-lattice-based digital signature standard. Technical Report FIPS 204, U.S. Department of Commerce, 2024.

[13] Phong Q. Nguyen and Oded Regev. Learning a parallelepiped: Cryptanalysis of GGH and NTRU signatures. In *Advances in Cryptology – EUROCRYPT 2006*, pages 271–288, 2006.

[14] Chris Peikert. A decade of lattice cryptography. *Foundations and Trends in Theoretical Computer Science*, 10(4):283–424, 2016.

[15] Oded Regev. On lattices, learning with errors, random linear codes, and cryptography. *Journal of the ACM*, 56(6):1–40, 2009. Preliminary version in STOC 2005.